

Isaac Henri Joël Houngue

Software and Research Engineer

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Summary

Motivated and hardworking Software and Research Engineer with MSc backgrounds in Artificial Intelligence and Computer Science, combining 3+ years of software engineering experience with research engineering experience in machine learning and scientific AI systems. Strong debugging and problem-solving skills; passionate about developing intelligent systems and translating research into usable, simple, effective, robust, and innovative solutions. Proactive and initiative-driven, accustomed to quality-focused environments and ready to take on challenging tasks with confidence and commitment.

Work Experience

Apprentice Research Engineer, InstaDeep – South Africa, hybrid Jul 2025 – Now

- Evaluated fine-tuning strategies for adapting InstaNovo, a transformer-based de novo peptide sequencing model, to glycopeptide sequencing tasks.
- Explored research directions including contrastive learning, multi-task learning, multi-view learning, knowledge distillation, curriculum learning, continual learning and transformer-variant architectures to improve sequencing performance and model generalization.
- Designed modular and pluggable components inspired by research papers to accelerate hypothesis testing and integration of new modeling approaches.
- Conducted large-scale experiments and ablation studies to measure the impact of architectural choices, training strategies, and model components on learning performance and generalization.
- Designed synthetic glycopeptide datasets and experimented with multi-stage training strategies to improve representation learning and transfer to experimental sequencing tasks.
- Performed glycopeptide spectral dataset analysis and feature selection studies to better understand data characteristics, model limitations, and sequencing behavior.
- Documented and communicated research findings through technical reports, visualizations, and presentations to guide future experimentation and development.

Backend Software Engineer, Fasfox – Paris, remote Mar 2023 – Sep 2024

- Enhanced the backend (API) of Concrete Dispatch SaaS, a logistics platform for managing concrete deliveries, by fixing bugs and adding features to support complex order and dispatch workflows (Django/DRF).
- Improved a fiber ordering and tracking product developed for AXIONE by adding features and fixing bugs related to fiber service management (Django/DRF).
- Developed a fault-tolerant daemon for parsing large CSV and eligibility data files used in Fiber Dispatch, a fiber ordering SaaS platform (Django/Ninja).
- Built TMForum compliant endpoints for fiber order handling and management following interop protocols (Django/Ninja).
- Retrieved information from web pages through scraping for real-time validation, and applied AI-powered OCR to extract data from PDF documents using external APIs (Django/DRF).
- Added features and maintained a daemon that listens to multiple email inboxes and parses email contents and attachments to automatically create objects in multiple databases (Django/DRF).

Frontend Software Engineer, Fasfox – Paris, remote Oct 2021 – Feb 2023

- Maintained and enhanced the Concrete Dispatch SaaS PWAs by implementing features such as photo uploads and management, digital signatures, QR code scanning, and interactive graphs, while ensuring high-quality code (Vue.js/Nuxt.js).
- Fixed bugs and developed an advanced slider for seamless concrete volume tracking in the Concrete Dispatch desktop web app (React.js/Next.js).
- Addressed bugs and added an efficient concrete calculator to the Concrete Dispatch blog site, aiming to attract more

users to the platform (Hugo/JavaScript).

Software Engineering Intern, National Civil Aviation Agency – Benin, onsite

Aug 2019 – Sep 2019

- Developed a Python-based inventory management desktop application using Tkinter to streamline the tracking of electronic equipment (Python/Tkinter).
- Assisted in the maintenance of office and IT equipment, including cleaning and basic upkeep tasks.

Personal Projects

Run Crucible 

Feb 2026 – Now

- Developed a modular framework for managing and comparing machine learning experiments across different training configurations and model architectures.
- Tools used: Python, PyTorch, Hydra, Docker

Recommender System 

Oct 2024 – Feb 2025

- Developed a recommender system using ALS and collaborative filtering built using the MovieLens dataset.
- Tools used: Python, NumPy, Streamlit

NSequence 

Jan 2024 – Feb 2024

- Developed a Python library for manipulating progressions or sequences.
- Tools used: Python

BVote 

Jun 2021 – Jul 2021

- Developed a secure voting platform with anonymity via blind signatures and blockchain for transparency.
- Tools used: Node.js, ReactJS, GraphQL, HyperLedger, MongoDB

Stream Processing Engine 

Feb 2021 – Mar 2021

- Built a simple e-commerce stream processing engine.
- Tools used: Python, Kafka, Redis, Socket.io

Billvoicer [Private]¹

Jan 2023 – Now

¹Private repository, available for demonstration

- Initiator and principal engineer of Billvoicer, driving the project's vision and technical development.
- Leading architecture design, feature development, and technology decisions.
- Managing a team of interns, overseeing development activities, issue tracking, code reviews and more.
- Ensuring seamless team communication and alignment with the product roadmap.
- Tools used: Python (Django/Drf), VueJS/NuxtJS, PostgreSQL, Redis, Celery

Technologies

Programming Languages: Python, JavaScript/TypeScript

Frameworks and libraries: Django, DRF, Flask, Ninja, Node.js, Vue.js, React.js, Next.js, GraphQL, PyTorch, Keras

Infrastructure and Tools: Linux, Git, Docker, GCP, AWS, Kafka, Spark, PostgreSQL, MongoDB, Neo4j

Education

Master in Artificial Intelligence, African Institute for Mathematical Sciences (AIMS), South Africa

Sep 2024 – Jul 2025

- **Coursework:** Bayesian Inference; Machine Learning at Scale; Active Learning, Information Theory and Uncertainty; Computer Vision; Reinforcement Learning; Deep Generative models and Generative AI; Engineering Large Language Models; CUDA Programming for GPUs and more.
- **Thesis:** Advancing De Novo Glycopeptide Sequencing with InstaNovo in Glycoproteomics.

Master in Computer Science, Institute of Mathematics and Physical Sciences (IMSP), Benin

Oct 2018 – Jul 2020

- **Coursework:** Algebra; Mathematical Analysis; Computer Architecture; Advanced Data Structure and Algorithms; Operating Systems; Software Architecture; Distributed Databases; Information and Communication Theory; Artificial Intelligence; Machine Learning and more.

- **Thesis:** A Smart Contract-based Remote Voting Platform using Blind Signature.

Languages

English: Professional proficiency

French: Native

Referees

Lélio Renard Lavaud 

Head of Engineering at Mistral AI, CEO and Chairman at Fasfox

Dr. Anicet Hounkanrin 

PhD in Electrical Engineering, University of Cape Town